Monthly Report for ECE FYP/FYT

Project Code:	YH02	Supervisor(s):	Zhang Fumin
Project Title:	Development of a Multi-Camera Vision-Based Localization System for Tracking Multiple Micro Autonomous Underwater Vehicles (AUVs)		
Group Member(s):	1) CHAN, Sze Sen 2) XU, Borong		
Reporting Period:	Report #1 ☐ Oct (Fall) Report #2 ☐ Nov (Fall) Report #3 ☑ Feb (Spring) (please attach Reports #1-2 to the Progress Report to be submitted in Jan) (please attach Reports #3 to the Final Report to be submitted in Apr)		
Progress Report: ● List the work completed in this reporting period. ● Identify the major difficulties encountered. ● Comment on the overall progress.	Work completed: ● Designed RSSI-based localization method. ● Investigated the relationship between RSSI value and distance between signal emitter and receiver. ● Investigate the calibration algorithm in OpenCV to apply the non-linear calibration. Major difficulties: ● The distance estimation has significant errors. The setup of data collection has to be modified. ● Signal from the emitter is unstable. Experiment environment or choice of signal emitter has to be adjusted. ● The OpenCV calibration algorithm includes a backward mapping and an updated intrinsic parameter for localization, major modification should be made for the non-linear calibration algorithm. Overall progress: ● Current progress is matching our goals.		
Future Plan: ● Write down the working plan for the next reporting period.	● Increase the amount and range of signal data collection. ● More review on the causes of signal instability. ● Future investigation and detailed analysis of OpenCV camera calibration algorithms. ● Combine the non-linear calibration algorithm with the localization algorithm.		
Group Representative's Signature:			

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